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Cutting Context Right: Chunking Strategies and Cross-Encoder Reranking for RAG

Wagner Nils* Wenzel Nick†

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Submitted to Carsten Lanquillon

*212405, nwagner1@stud.hs-heilbronn.de
†207678, nwenzel@stud.hs-heilbronn.de

1 Abstract

This report investigates how chunking strategy, chunk size, overlap, and cross-encoder reranking influence retrieval quality and answer accuracy in a Retrieval-Augmented Generation pipeline. The experiment uses the Wikipedia AI/DL corpus with 20 documents and 40 evaluation questions. We compare fixed-size character splitting, sentence-based splitting, and recursive character splitting with chunk sizes of 128, 256, 512, and 1024 characters and overlap settings of 0 and 50 characters. Dense retrieval uses `all-MiniLM-L6-v2` embeddings with a normalized `IndexFlatIP` FAISS index. Reranking is evaluated by retrieving the top-20 bi-encoder candidates and reranking them to the final top-5 with `cross-encoder/ms-marco-MiniLM-L-6-v2`. The strongest bi-encoder-only configuration is sentence-based splitting with size 128 and overlap 0 or 50, reaching Recall@5 = 1.000 and MRR = 0.946. The best reranked configuration is sentence-based splitting with size 512 and no overlap, reaching Recall@5 = 1.000 and MRR@5 = 0.971. Answer accuracy was evaluated with `google/flan-t5-small` for three representative configurations and reached 0.40, 0.05, and 0.00, showing that strong retrieval alone does not guarantee strong generation. Practically, sentence-based medium chunks with reranking are recommended when retrieval precision is most important.

2 Introduction

Retrieval-Augmented Generation (RAG) connects a language model to external knowledge without changing the model weights [1, 3]. In such a system, the retrieval stage is a central bottleneck: if the relevant evidence is not retrieved, the generator cannot reliably produce a grounded answer. Retrieval quality is therefore not only an implementation concern, but a core modeling decision.

One important retrieval decision is how the source documents are split into chunks. If chunks are too small, they may lose the surrounding context needed to answer a question. If chunks are too large, the embedding vector must represent too much heterogeneous information, which can dilute the semantic signal. The splitting method also matters. A fixed-size splitter is simple and predictable, but it may cut through sentences. Sentence-based and recursive splitters preserve more linguistic or structural boundaries and may therefore produce more meaningful retrieval units. Overlap can further reduce boundary problems, but it also increases the number of chunks and may introduce redundant candidates.

Cross-encoder reranking adds a second retrieval stage after fast bi-encoder search. A bi-encoder embeds queries and chunks independently, which enables efficient vector search with FAISS. However, it cannot model detailed token-level interactions between the query and a candidate chunk [2]. A cross-encoder jointly scores a query–chunk pair and can therefore refine the ranking, but it is too slow to apply to the full corpus. The two-stage pattern retrieves a candidate pool with the bi-encoder and reranks only this smaller set.

The research question is: how do chunking strategy, chunk size, overlap, and cross-encoder reranking influence retrieval quality and answer accuracy of a RAG pipeline? We expect medium-sized chunks to work best, overlap to help mainly when relevant information lies near chunk boundaries, and reranking to improve top-5 ranking quality when the relevant chunk is already present in the bi-encoder candidate pool.

3 Methods

3.1 Corpus and evaluation setup

The experiment uses the Wikipedia AI/DL corpus provided for the assignment. It contains 20 Wikipedia articles with a total of 110,494 words. The evaluation set contains 40 questions: 15 easy, 15 medium, and 10 hard. Each question includes an expected answer, a difficulty label, and a gold source article. In the notebook implementation, a retrieved chunk is counted as relevant if it comes from the same gold source article as the answer. This means that retrieval relevance is evaluated at source-article level rather than exact evidence-span level.

3.2 Chunking configurations

Three chunking methods are compared. The fixed-size method splits documents into consecutive character windows. The sentence-based method uses NLTK sentence tokenization and groups complete sentences until the target character size is reached. The recursive method tries paragraph, newline, sentence, and word boundaries and is conceptually similar to recursive character splitting used in common RAG pipelines.

For each method, the experiment evaluates four target chunk sizes,

$$\{128, 256, 512, 1024\},$$

and two overlap settings,

$$\{0, 50\}.$$

Both size and overlap are measured in characters, not tokens. This gives $3 \times 4 \times 2 = 24$ retrieval configurations. The number of resulting chunks differs strongly by method and size. For example, sentence-based splitting with size 512 and no overlap produces 1775 chunks, while recursive splitting with size 128 and overlap 50 produces 12288 chunks.

3.3 Embedding, indexing, retrieval, and reranking

All configurations use the same embedding model, `all-MiniLM-L6-v2`, with embedding dimension 384. Chunk embeddings and query embeddings are normalized. FAISS indexing uses `IndexFlatIP`; with normalized vectors, inner product corresponds to cosine similarity. For each configuration, the system retrieves the top-20 chunks with the bi-encoder. Retrieval quality is evaluated at Recall@5, Recall@10, and MRR.

Cross-encoder reranking is evaluated for every chunking configuration. The reranker is `cross-encoder/ms-marco-MiniLM-L-6-v2`. For each query, the top-20 FAISS candidates are scored jointly with the query, sorted by cross-encoder score, and truncated to the final top-5. This isolates the effect of reranking while keeping the candidate pool fixed.

Answer accuracy is implemented only for three representative configurations, not for the full grid. The generator is `google/flan-t5-small`. The top-5 retrieved chunks are concatenated into a context of at most 3500 characters. Generated answers are compared to the gold answer using an approximate automatic match based on normalized substring matching and `SequenceMatcher` similarity with a threshold of 0.45. Therefore, answer accuracy should be interpreted as a coarse sanity check rather than a human-quality evaluation.

4 Results

4.1 Bi-encoder retrieval

Table 1 summarizes the strongest configurations. Several configurations reach perfect Recall@5 and Recall@10, so MRR is the more discriminating metric.

Criterion	Configuration	Recall@5	Recall@10	MRR/MRR@5	
Best bi-encoder	sentence, 128, overlap 0/50	1.000	1.000	0.946	0.015–0.019s
Best reranked	sentence, 512, overlap 0	1.000	–	0.971	0.57s
Best fixed-size bi-encoder	fixed, 512, overlap 50	1.000	1.000	0.944	0.014s
Best recursive bi-encoder	recursive, 128, overlap 50	1.000	1.000	0.933	0.023s

Table 1: Best observed configurations. Chunk sizes and overlaps are measured in characters. Reranked results report final top-5 performance after reranking top-20 FAISS candidates.

The best bi-encoder-only configuration is sentence-based splitting with size 128 and either no overlap or 50 characters overlap. Both variants reach Recall@5 = 1.000, Recall@10 = 1.000, and MRR = 0.946. Sentence-based splitting with size 512 is also strong, reaching Recall@5 = 1.000 and MRR around 0.936–0.938 before reranking.

Chunk size has a visible but not fully monotonic effect. With no overlap, the plotted results show that size 128 and size 512 are particularly strong for sentence-based and recursive splitting, while size 1024 is less reliable. Fixed-size splitting is less stable and only reaches perfect Recall@5 in the 512-character setting with overlap. This supports the expectation that very large chunks can dilute the embedding signal, but it does not show that the largest possible context is always best.

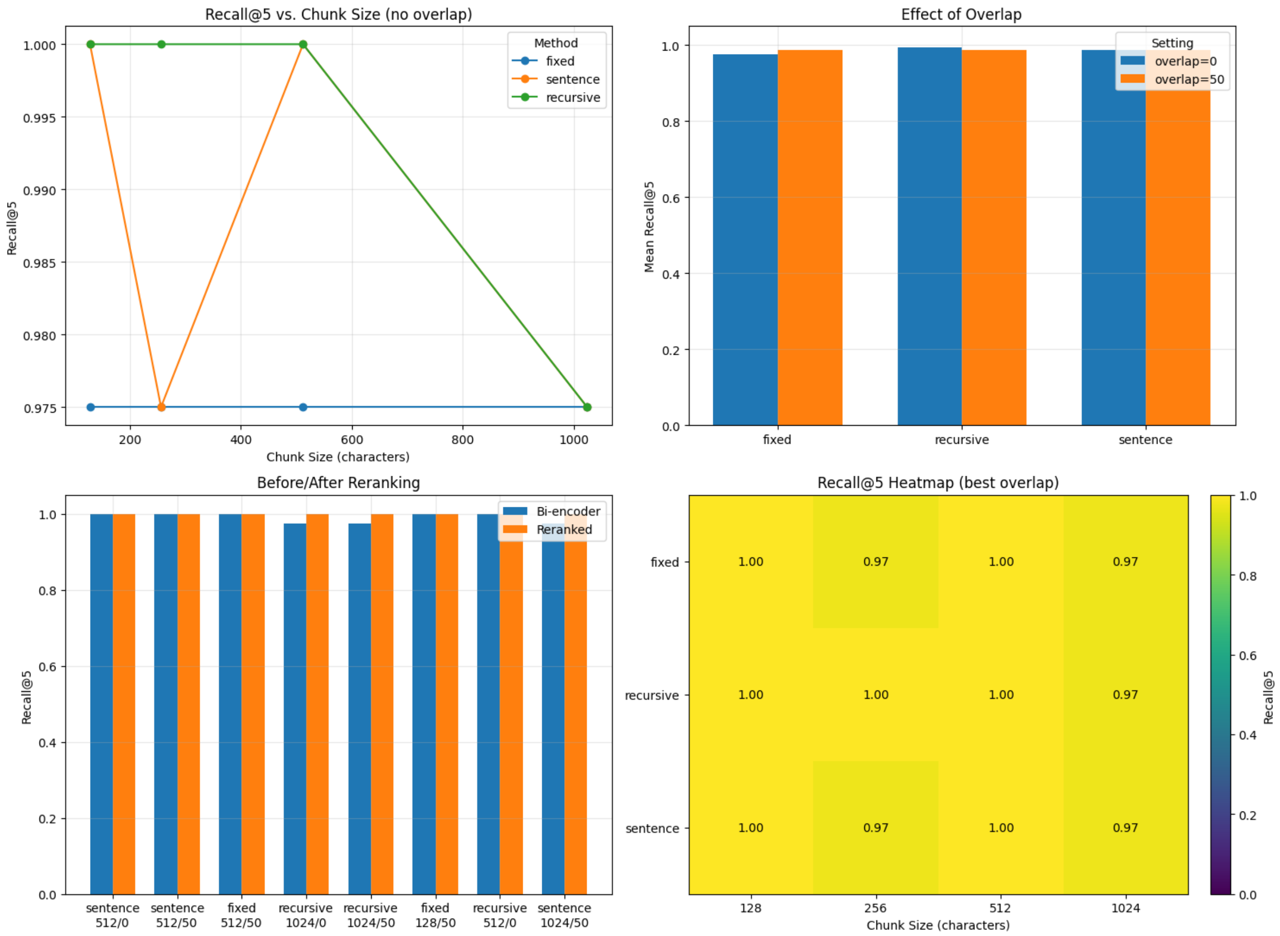

Figure 1: Compact overview of retrieval results. (a) Recall@5 by chunk size without overlap. (b) Mean Recall@5 for overlap 0 and overlap 50 by chunking method. (c) Recall@5 before and after cross-encoder reranking for selected configurations. (d) Recall@5 heatmap for the best overlap setting across chunking methods and chunk sizes.

The heatmap in Figure 1 should be interpreted with caution because Recall@5 is already saturated for many configurations. Most method-size combinations retrieve at least one chunk from the correct source article in the top five, resulting in values close to 1.00. Therefore, the heatmap is mainly useful as a compact overview of robustness across chunking settings, while MRR provides a more discriminating comparison between strong configurations. The limited variation in the heatmap is also influenced by the article-level relevance definition: a retrieved chunk is counted as relevant if it comes from the correct gold source article, even if it does not necessarily contain the exact evidence span.

4.2 Effect of overlap

Overlap has only a marginal overall effect. The overlap plot in Figure 1 shows that overlap 50 slightly improves fixed-size splitting on average, while the effect for sentence-based and recursive splitting is very small. This is consistent with the idea that overlap mainly helps when splitting can cut through relevant information. Since sentence-based splitting already preserves complete sentences, overlap is less decisive there. The best reranked configuration uses no overlap.

4.3 Effect of reranking

Cross-encoder reranking improves ranking quality for the best overall configuration. Sentence-based splitting with size 512 and no overlap improves from MRR@5 = 0.936 before reranking to MRR@5 = 0.971 after reranking, while Recall@5 remains 1.000. The same MRR@5 is reached by the size 512, overlap 50 sentence-based configuration, but the no-overlap version is selected as the best configuration in the notebook.

Reranking is not uniformly beneficial. Some configurations improve substantially, for example fixed-size splitting with size 128 and no overlap improves from Recall@5 = 0.975 to 1.000 and from MRR@5 = 0.877 to 0.921. However, reranking can also reduce performance: recursive splitting with size 128 drops from Recall@5 = 1.000 to 0.975 after reranking. Thus, reranking helps when the correct document is present in the candidate pool and the cross-encoder promotes it, but it can hurt when it displaces an already relevant top-5 candidate.

The computational cost is substantial. For the best reranked configuration, reranking takes about 0.57 seconds for the 40 evaluation queries, while bi-encoder query time is about 0.013 seconds. This means that the reranking stage is tens of times slower than dense retrieval alone in this small experiment.

4.4 Answer accuracy

Answer accuracy was evaluated only for three representative configurations. The results are shown in Table 2. The strongest bi-encoder retrieval setup gives the highest generated-answer accuracy, while the reranked setup performs much worse in generation despite excellent retrieval metrics.

Configuration	Answer accuracy	Questions	Correct
sentence, 128, overlap 0, bi-encoder	0.40	40	16
sentence, 512, overlap 0, reranked	0.05	40	2
fixed, 1024, overlap 0, bi-encoder	0.00	40	0

Table 2: Answer accuracy for three representative configurations using `google/flan-t5-small`. The automatic evaluation is approximate and was not run for the full experimental grid.

These results show that high retrieval quality is necessary but not sufficient for high answer accuracy. The small generator often produced short, incomplete, or poorly aligned answers. For example, the answer to the AI founding question is predicted as “1956” and counted as correct, while some other outputs contain artifacts such as chunk labels or overly compressed answers. Therefore, the retrieval metrics remain the primary basis for comparing chunking and reranking, while answer accuracy mainly highlights limitations of the generation stage.

5 Discussion

The best overall retrieval configuration is sentence-based chunking with size 512, no overlap, and cross-encoder reranking. It reaches Recall@5 = 1.000 and the highest observed MRR@5 of 0.971. This supports the idea that preserving sentence boundaries is useful for RAG retrieval. The chunks remain semantically coherent while still being compact enough for the embedding model and informative enough for the cross-encoder.

The best bi-encoder-only configuration is also sentence-based, but with size 128 and overlap 0 or 50. It reaches Recall@5 = 1.000 and MRR = 0.946 without the reranking stage. This suggests that small sentence-preserving chunks can work very well when the evaluation target is source-article retrieval. However, the answer accuracy experiment shows that small chunks do not automatically guarantee strong final answers, because generation quality depends on how well the retrieved context can be synthesized by the model.

The expected chunk-size sweet spot is partly visible. Very large chunks of 1024 characters are weaker in several settings, which is consistent with diluted embedding representations. However, the strongest direct retrieval result occurs at 128 characters, while the strongest reranked result occurs at 512 characters. This indicates an interaction between chunk size and reranking: small chunks may be enough for bi-encoder source matching, whereas medium chunks provide the cross-encoder and generator with richer context.

Overlap helps less than expected. It improves some fixed-size settings, which is plausible because fixed windows can cut through relevant information. However, overlap is not decisive for sentence-based chunking and is not needed for the best reranked configuration. In this corpus, respecting sentence boundaries appears more important than adding fixed character overlap.

Sentence-based and recursive splitting both outperform fixed-size splitting in the strongest configurations. Recursive splitting is competitive, but it does not clearly beat sentence-based splitting in this notebook. The best single configuration is sentence-based, and the best fixed-size setting requires overlap to match the perfect Recall@5 of the better boundary-aware methods.

The best configuration has no top-5 retrieval failures. Therefore, the notebook uses diagnostic failures from strong imperfect configurations. Five diagnostic failures are reported. Three are reranking regressions: for the ImageNet benchmark question, the gold document *Computer vision* was present before reranking but displaced from the final top-5 in recursive size 128 settings and fixed size 256. One failure is a lexical similarity trap: the Bard incident question should retrieve *Retrieval-augmented generation*, but retrieved chunks from *Artificial intelligence* and *Large language model*. One failure is classified as a boundary or representation miss, where the relevant document was absent from the top-20 candidate pool. These examples show that reranking can refine rankings, but it cannot recover evidence that is missing from the candidate pool and may occasionally demote relevant candidates.

The main limitations are the small corpus size, the use of only one embedding model, the use of only one cross-encoder reranker, and the article-level relevance definition. Since a chunk is relevant if it comes from the correct source article, the evaluation does not verify that the retrieved chunk contains the exact evidence sentence. Answer accuracy is also limited because it was evaluated only on three configurations with `google/flan-t5-small` and an approximate automatic matching rule. No human answer-quality evaluation was performed.

For a similar Wikipedia-style RAG corpus, the practical recommendation is to use sentence-based chunking with a medium chunk size around 512 characters, no overlap, top-20 bi-encoder retrieval, and cross-encoder reranking to the final top-5 when retrieval precision is more important than latency. If latency is the dominant constraint, sentence-based splitting with size 128 and no reranking is a strong fallback because it already reaches Recall@5 = 1.000 and MRR = 0.946 in this experiment.

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